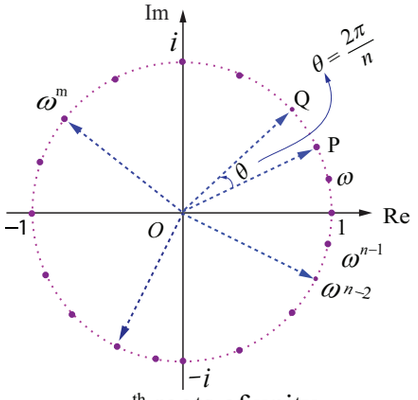
Therefore ω is an nth root of unity. From equation (1), the complex numbers $1, \omega, \omega^2, \cdots, \omega^{n-1}$ are n^{th} roots of unity. The complex numbers $1, \omega, \omega^2, \cdots, \omega^{n-1}$ are the points in the complex plane and are the vertices of a regular polygon of n sides inscribed in a unit circle as shown in Fig 2.45. Note that because the nth roots all have the same modulus 1, they will lie on a circle of radius 1 with centre at the origin. Furthermore, the n roots are equally spaced along the circle, because successive n^{th} roots have arguments that differ by $\frac{2\pi}{n}$.



n^{th} roots of unity

Fig. 2.45

The nth roots of unity $1, \omega, \omega^2, \cdots, \omega^{n-1}$ are in geometric progression with common ratio ω.

Therefore $1 + \omega + \omega^2 + \cdots + \omega^{n-1} = \dfrac{1-\omega^n}{1-\omega} = 0$ since $\omega^n = 1$ and $\omega \neq 1$.

The sum of all the nth roots of unity is
$1 + \omega + \omega^2 + \cdots + \omega^{n-1} = 0$.

The product of n, n^{th} roots of unit is

$$1\,\omega\,\omega^2 \cdots \omega^{n-1} = \omega^{0+1+2+3+\cdots+(n-1)} = \omega^{\frac{(n-1)n}{2}}$$

$$= \left(\omega^n\right)^{\frac{(n-1)}{2}} = \left(e^{i2\pi}\right)^{\frac{(n-1)}{2}} = \left(e^{i\pi}\right)^{n-1} = (-1)^{n-1}$$

The product of all the nth roots of unity is
$1\,\omega\,\omega^2 \cdots \omega^{n-1} = (-1)^{n-1}$.

Since $|\omega| = 1$, we have $\omega\bar{\omega} = |\omega|^2 = 1$; hence $\bar{\omega} = \omega^{-1} \Rightarrow (\bar{\omega})^k = \omega^{-k}$, $0 \le k \le n-1$

$$\omega^{n-k} = \omega^n \omega^{-k} = \omega^{-k} = (\bar{\omega})^k,\ 0 \le k \le n-1$$

Therefore, $\omega^{n-k} = \omega^{-k} = \left(\bar{\omega}\right)^k,\ 0 \le k \le n-1.$

Note

(1) All the n roots of nth roots unity are in Geometrical Progression

(2) Sum of the n roots of nth roots unity is always equal to zero.

(3) Product of the n roots of nth roots unity is equal to $(-1)^{n-1}$.

(4) All the n roots of nth roots unity lie on the circumference of a circle whose centre is at the origin and radius equal to 1 and these roots divide the circle into n equal parts and form a polygon of n sides.

12th_Maths_EM_Vol1_CH 2_Complex Numbers.indd 88 16-12-2022 12:30:25